

You can open settings on the Android phone as shown in Fig. 3. You can choose your preferred language as shown in Fig. 4.

Next, select the 'Routines' option from Google Assistant. You will get the '+ Add a routine' button from

Routines (Fig. 5). Click on it and set the voice command that you want to create (Fig. 6). For example, if you want to control lights, you can set the voice commands in your preferred language to turn on the light bulb. When you give a voice command, the



Fig. 11: Creating URL of Raspberry Pi server

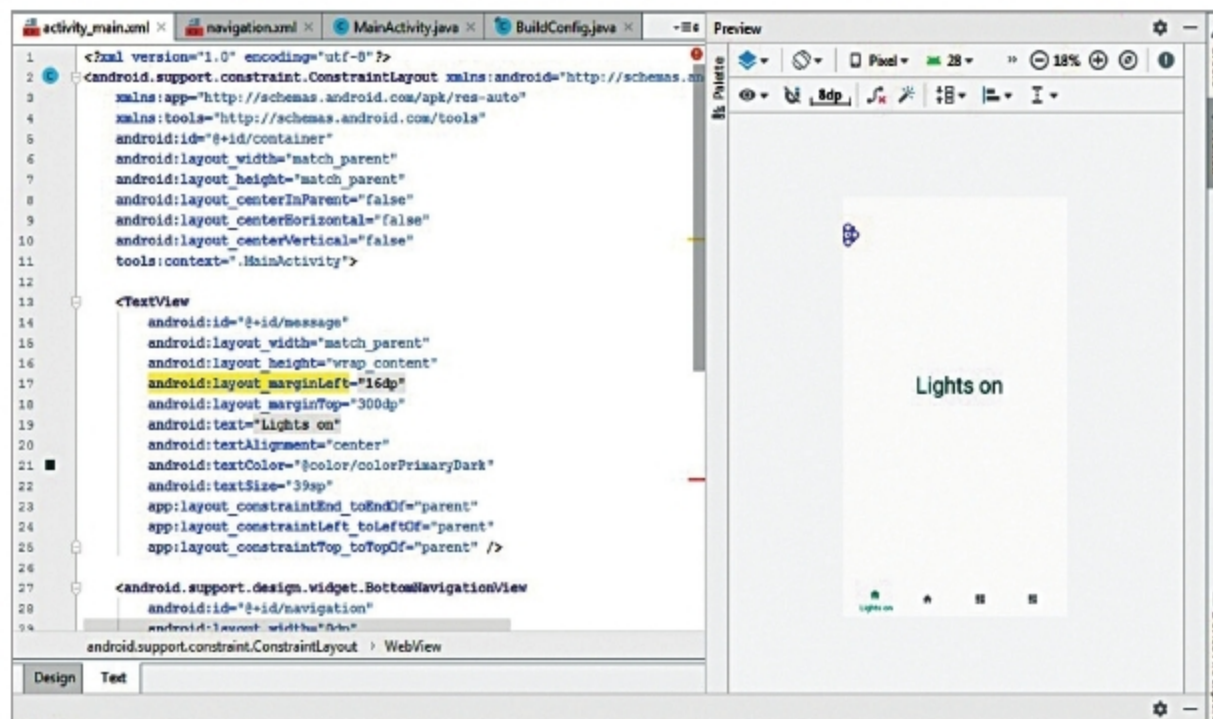


Fig. 12: Creating app layout

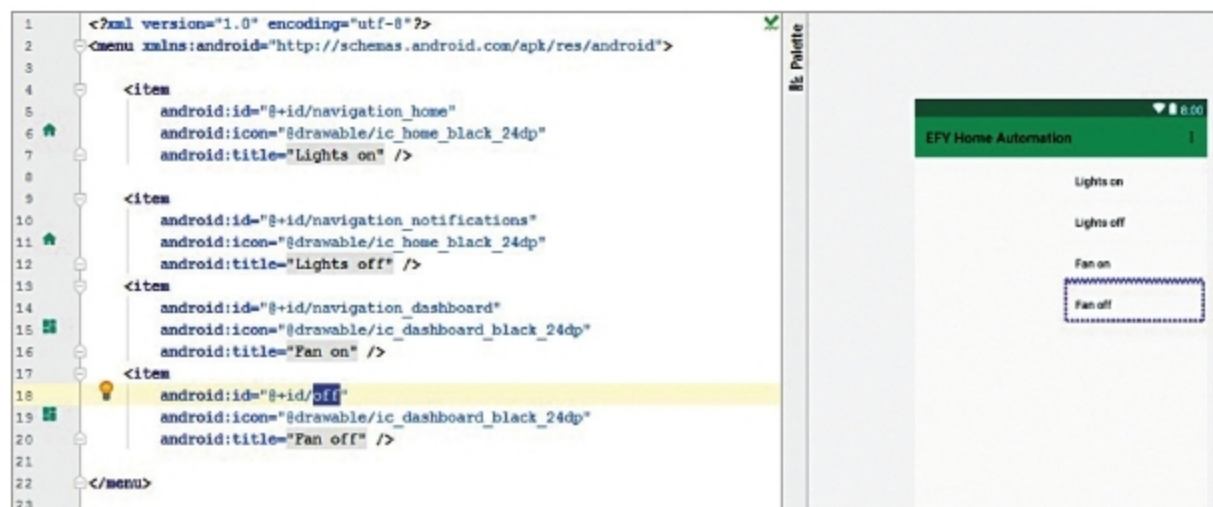


Fig. 13: Creating navigation bar

EFY Note

The source code of this project is available for free download at source.efymag.com

system will search for a Raspberry Pi IP address such as 192.168..... or PHP file name lightsoff.php.

Now, connect the components as shown in Fig. 7 and Table 2. Enclose the circuit in a suitable box as shown in Fig. 8.

Android app development

We can also create our own app to control the light and fan. Let's make the app using Android Studio IDE. You can download Android Studio from <https://developer.android.com/studio>.

Open Android Studio and create a new project with the bottom navigation bar.

First of all, set the permission in the code (Fig. 9) to access the Internet because we are going to use Wi-Fi to control the GPIO of Raspberry Pi.

Then go to MainActivity.java and create WebView and String variables as shown in Fig. 10.

After that, set the URL for Raspberry Pi as shown in Fig. 11.

Next, create a layout of the app as shown in Fig. 12. Save the project with a suitable name, say HomeAutomation.apk. Install this .apk file on your Android phone.

Testing

Power the Raspberry Pi with 5V DC. Then connect Raspberry Pi and your phone to your Wi-Fi network, or a phone's hotspot. Then say 'Hey Google' followed by the voice command that you have set in your preferred language. For example, say 'Hey Google light on kar do.' The system will turn on the light.

You can also control the light and fan with the Home Automation app you have just created. Open the app and press the icon in the navigation to turn the light on or off. Similarly, you can control the fan as well from the app. **EFY**

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